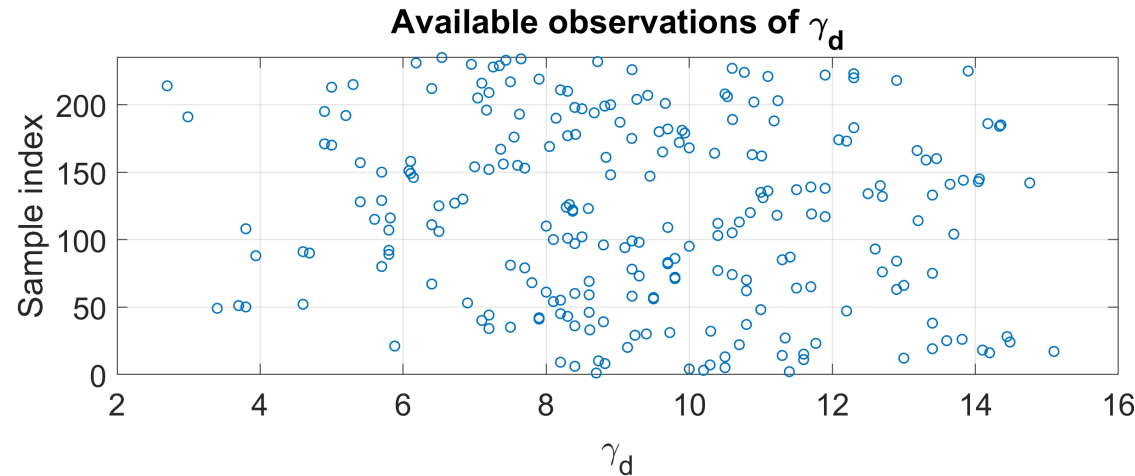


Modelling, Uncertainty and Data for Engineers (MUDE)

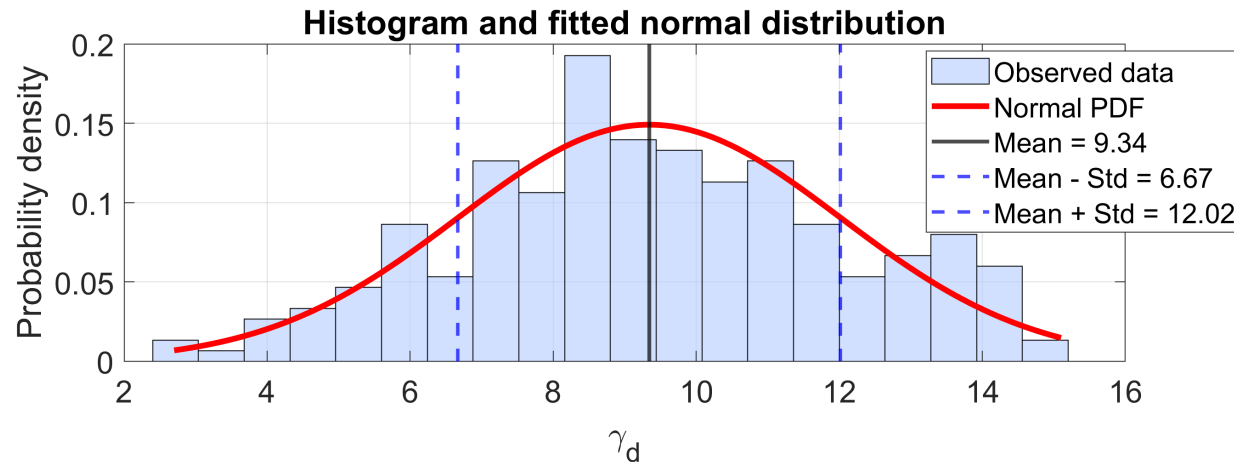
Week 1.5 : Multivariate distributions

Zheng Guan

Recap Week 1: Modelling a single variable

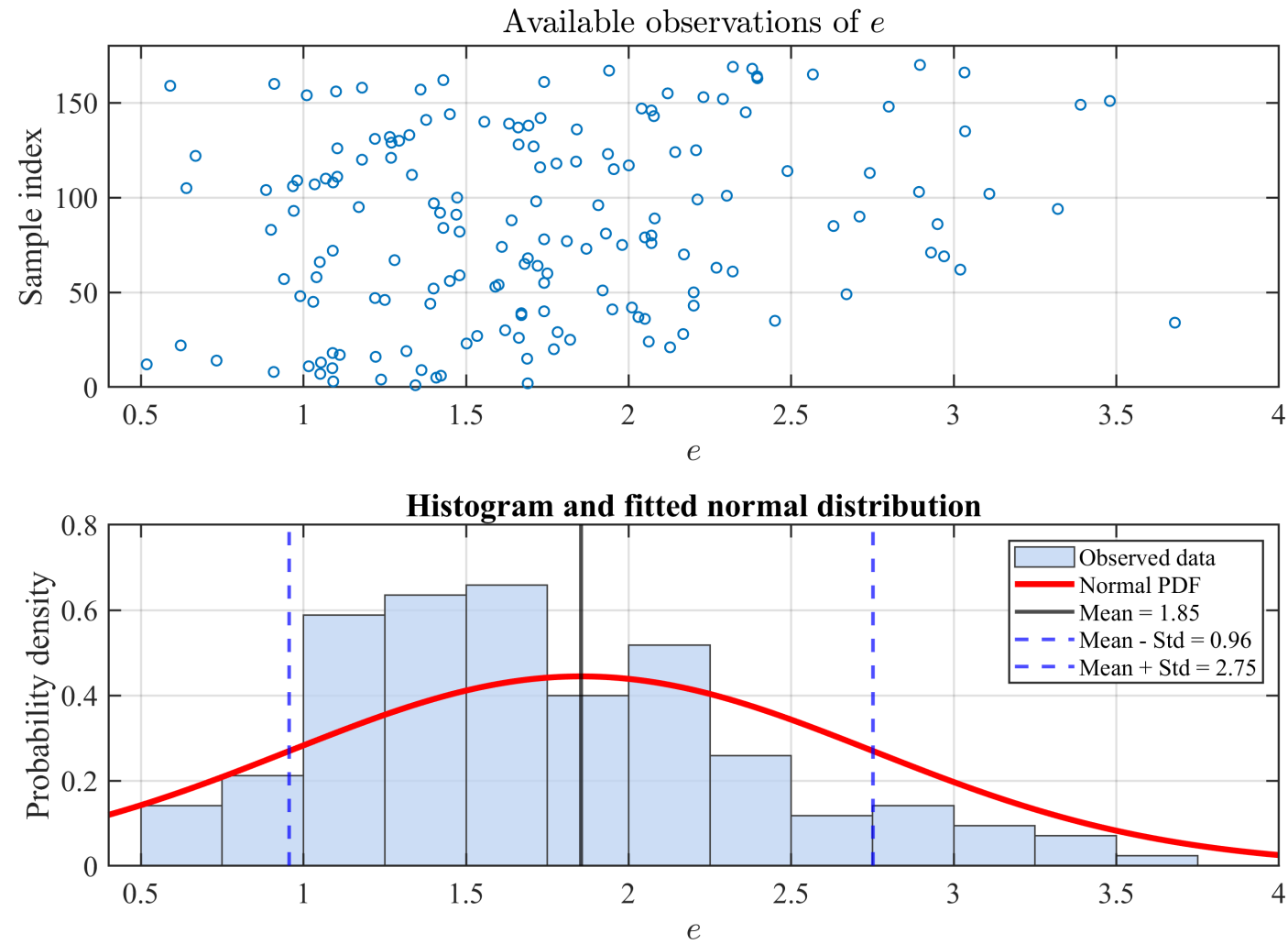


Measured **dry unit weight**, γ_d values of Dutch clay deposits

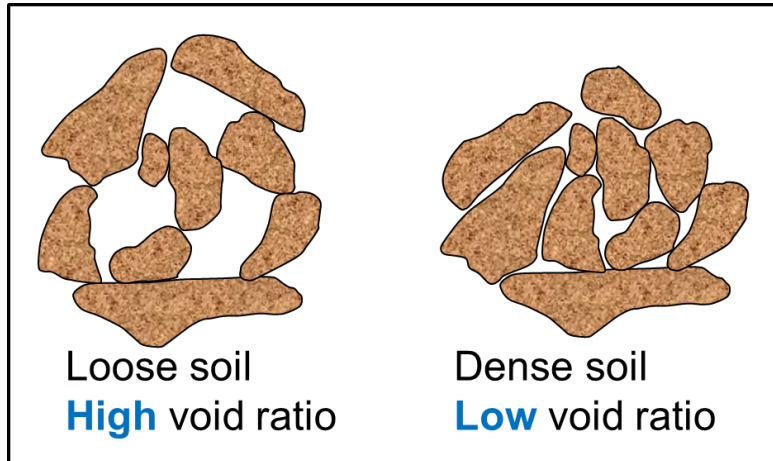


A single variable can often be represented using a probability distribution

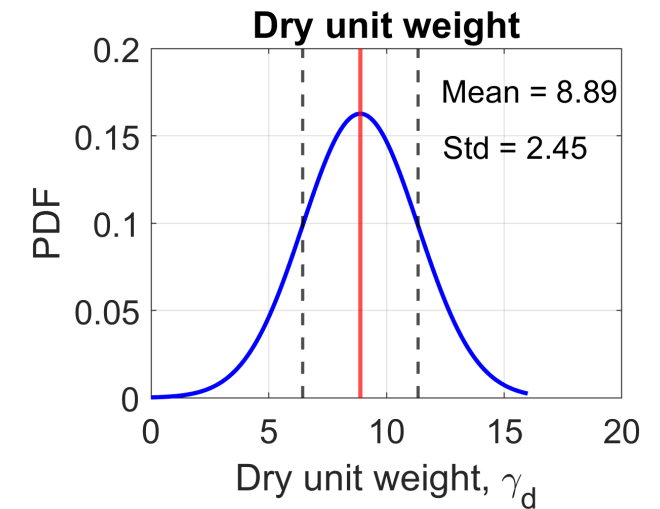
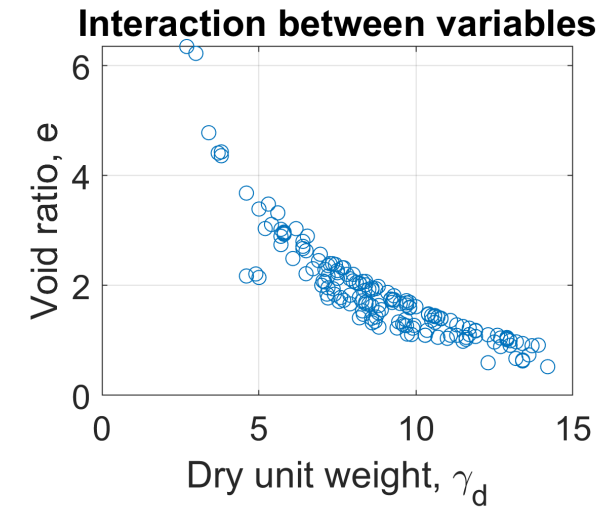
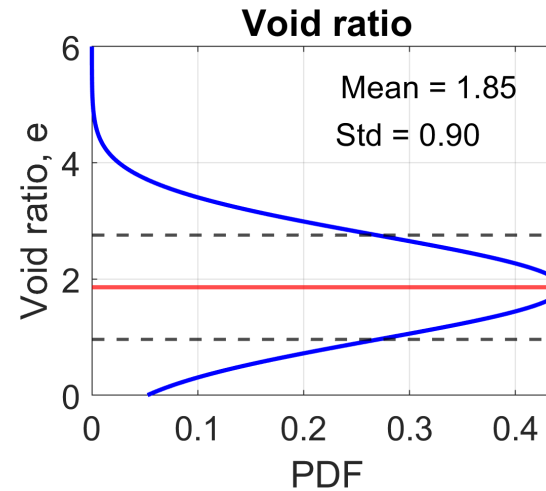
Another variable: Void ratio



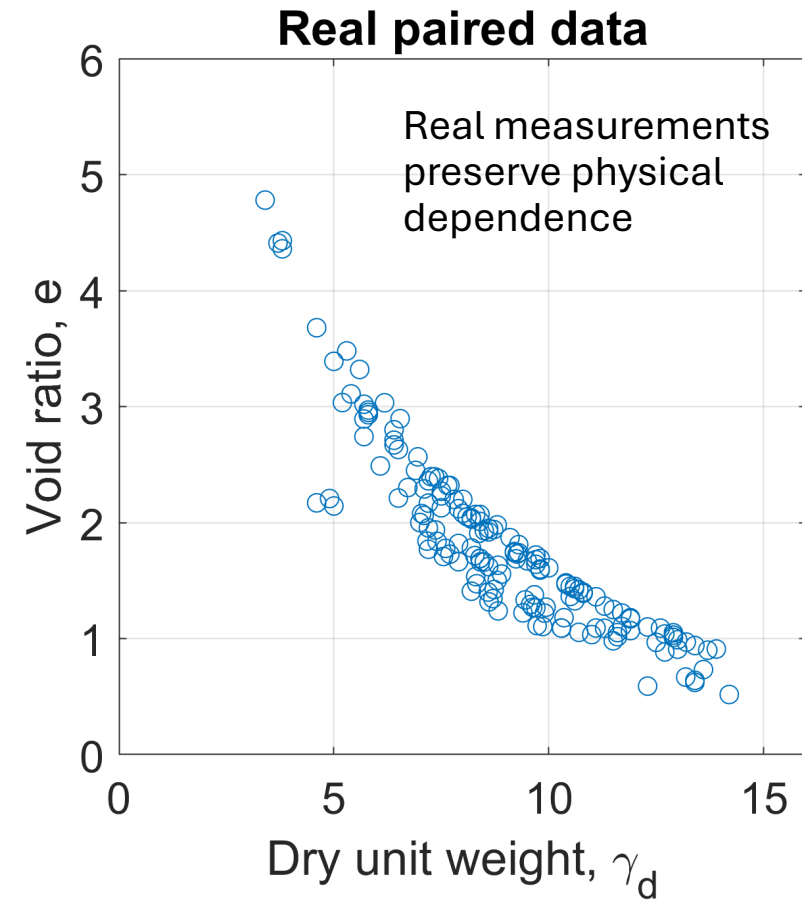
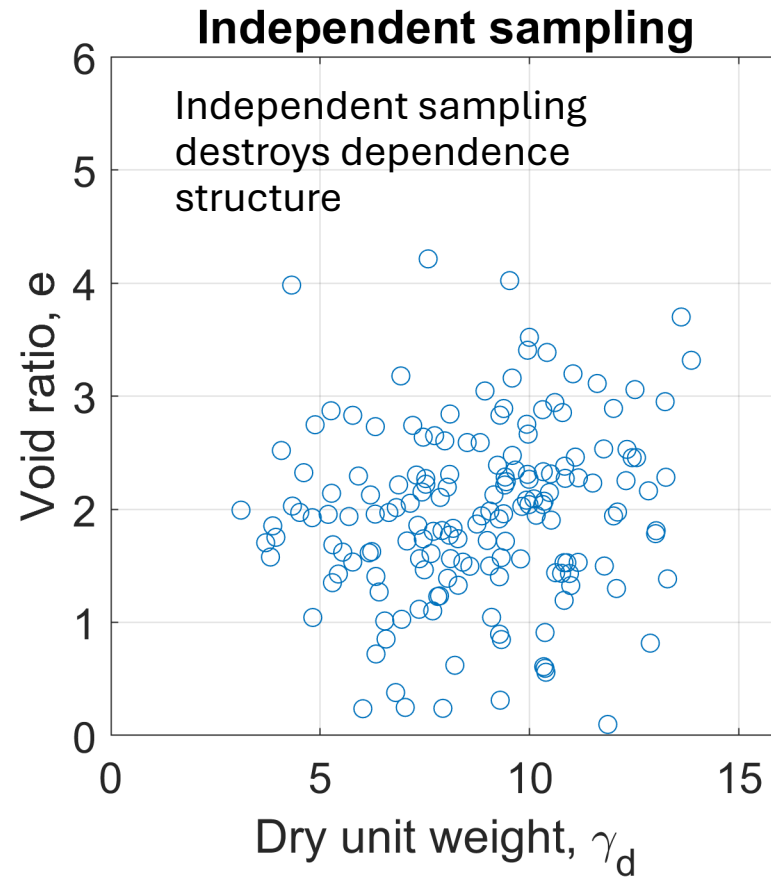
Interaction between variables



Loose soils tend to have higher void ratio and lower dry unit weight.



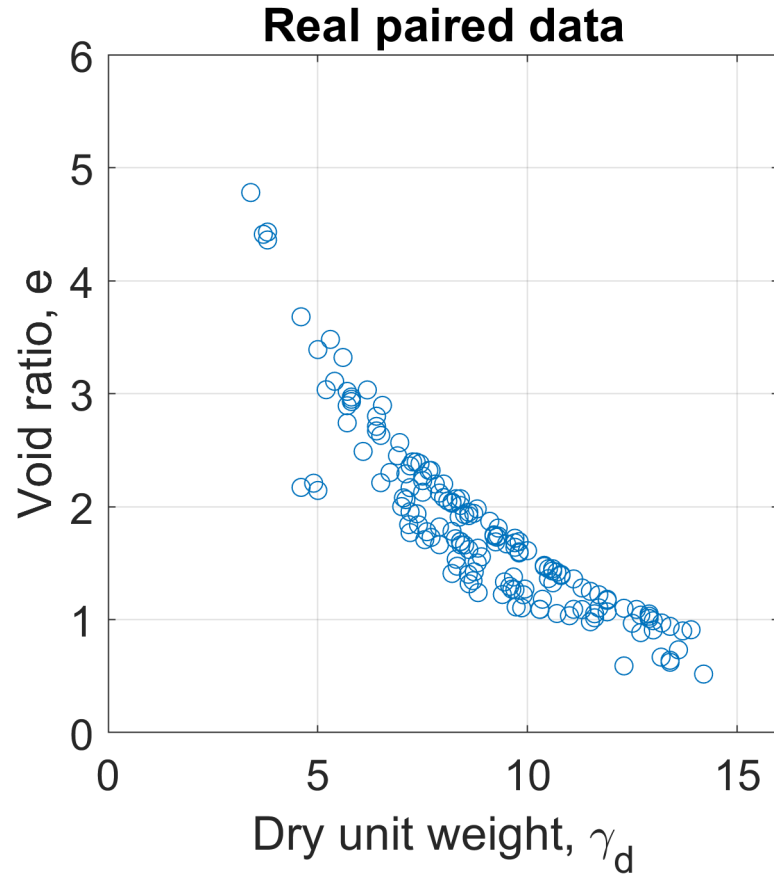
What happens if we model the variables independently?



Ignoring dependence loses important information

What about Week 5?

From one variable to multiple interacting variables



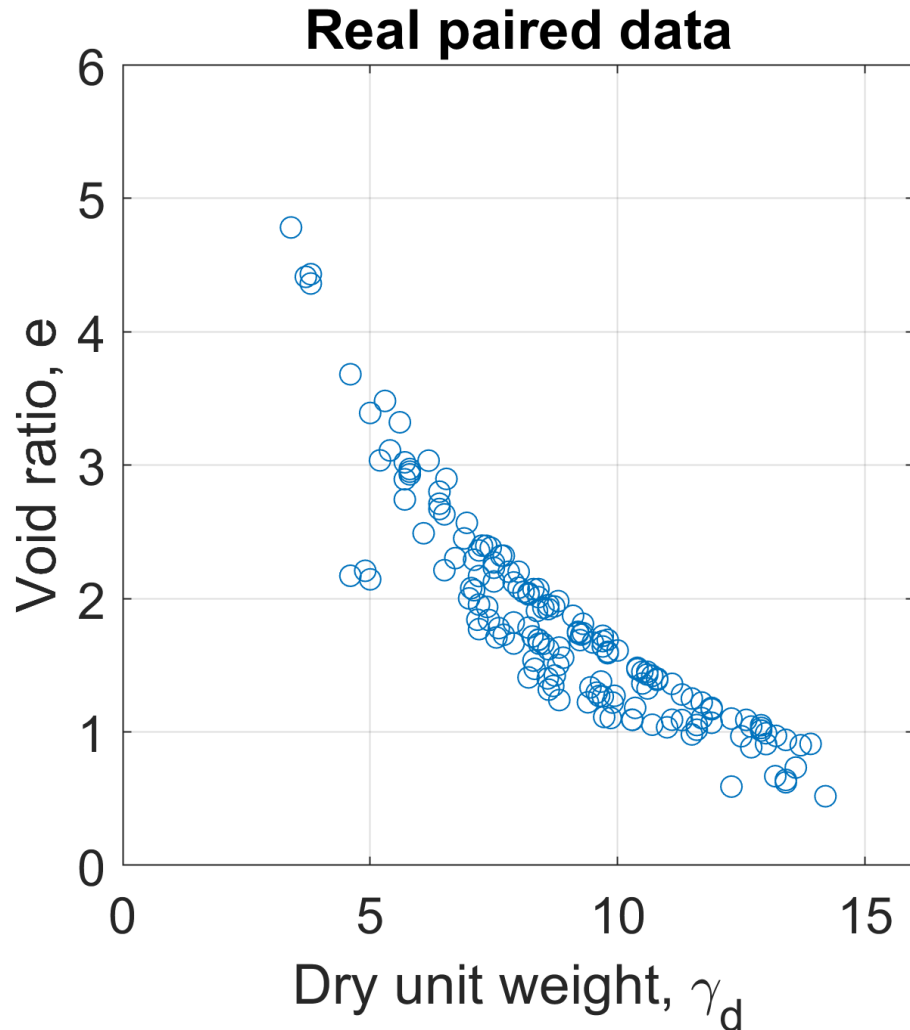
- Engineering parameters often interact
- We need joint probability models

Contents – Multivariate distributions

1. Measuring dependence (covariance/correlation)
2. Joint probability distribution
3. Bivariate Gaussian distribution
4. Multivariate Gaussian distribution

Measures of dependence

Measures of dependence



How are the two variables related?

Direction of dependence: positive or negative?

How strong is the relationship?

Strength of dependence: weak or strong?

Sample Covariance

$$Cov(X_1, X_2) = \frac{1}{n-1} \sum_{i=1}^n \underbrace{(X_{1,i} - \bar{X}_1)}_{\text{Deviation of } X_1} \underbrace{(X_{2,i} - \bar{X}_2)}_{\text{Deviation of } X_2}$$

Sample mean



where:

- $X_{1,i}$ = i -th observation of variable X_1
- $X_{2,i}$ = i -th observation of variable X_2
- $\bar{X}_1$ = mean of X_1
- $\bar{X}_2$ = mean of X_2
- n = number of observations

We use **$n-1$** because the true means are unknown and are estimated from the sample

Covariance

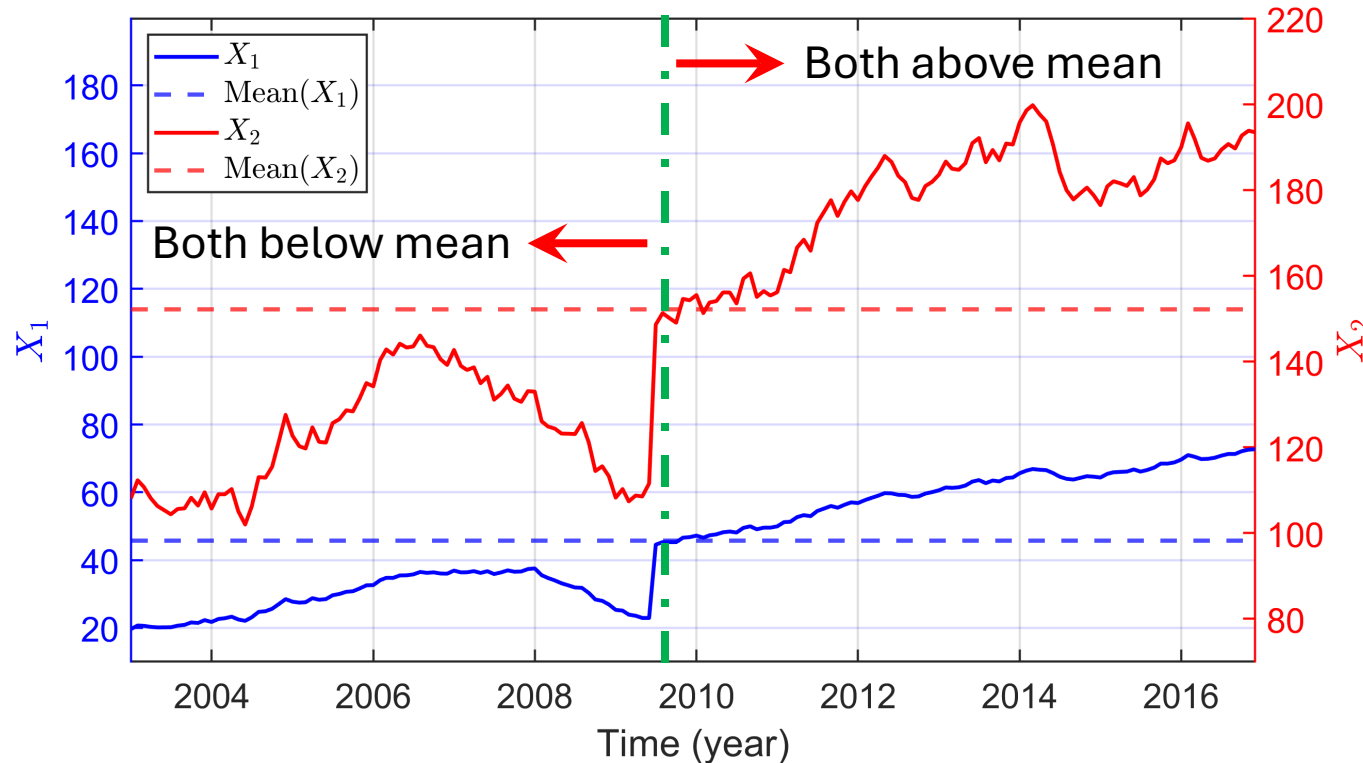
$$\text{Cov}(X_1, X_2) = \frac{1}{n-1} \sum_{i=1}^n \underbrace{(X_{1,i} - \bar{X}_1)}_{\text{Deviation of } X_1} \underbrace{(X_{2,i} - \bar{X}_2)}_{\text{Deviation of } X_2}$$

Case	$X_{1,i} - \bar{X}_1$	$X_{2,i} - \bar{X}_2$	Product	Covariance Contribution
1	+	+	+	Positive
2	-	-	+	Positive
3	-	+	-	Negative
4	+	-	-	Negative

Positive contribution
when variables tend to be above or below their means together

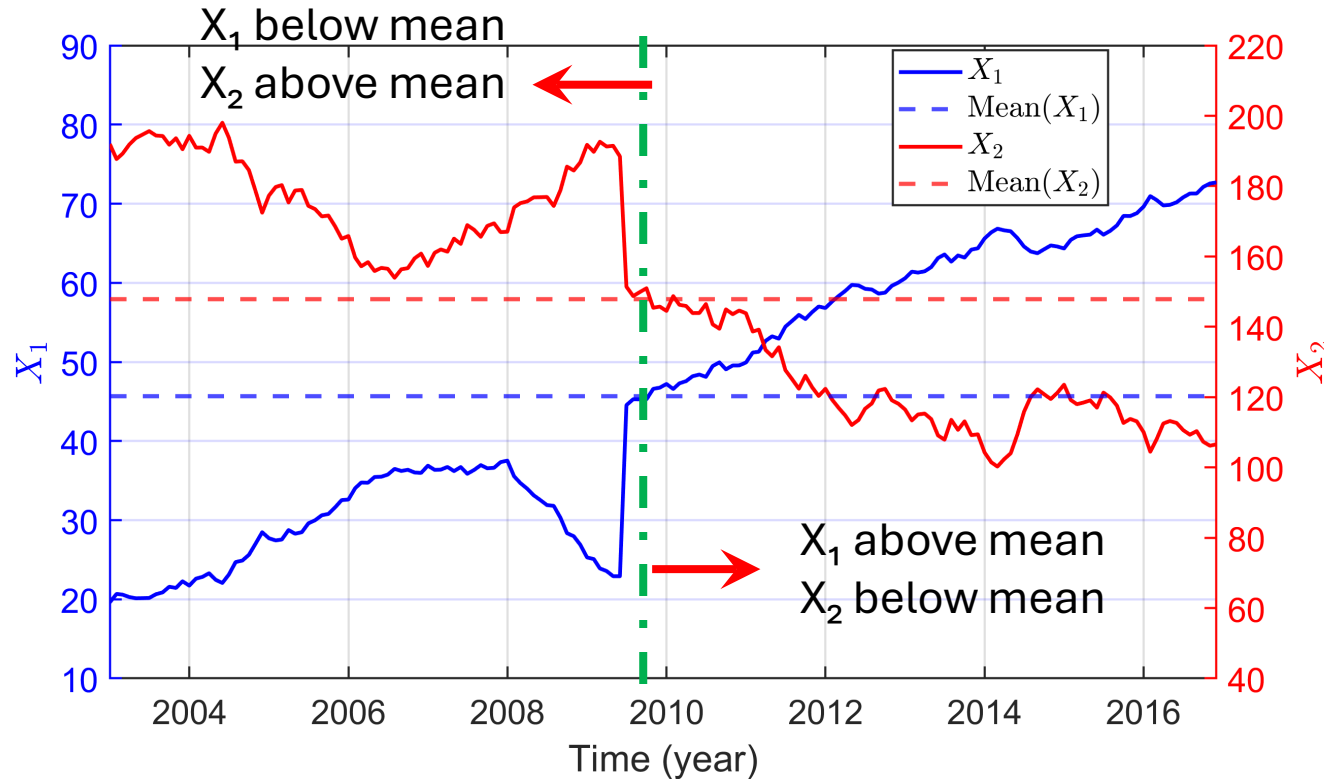
Negative contribution
when one tends to be above its mean while the other is below its mean¹¹

Geometric interpretation



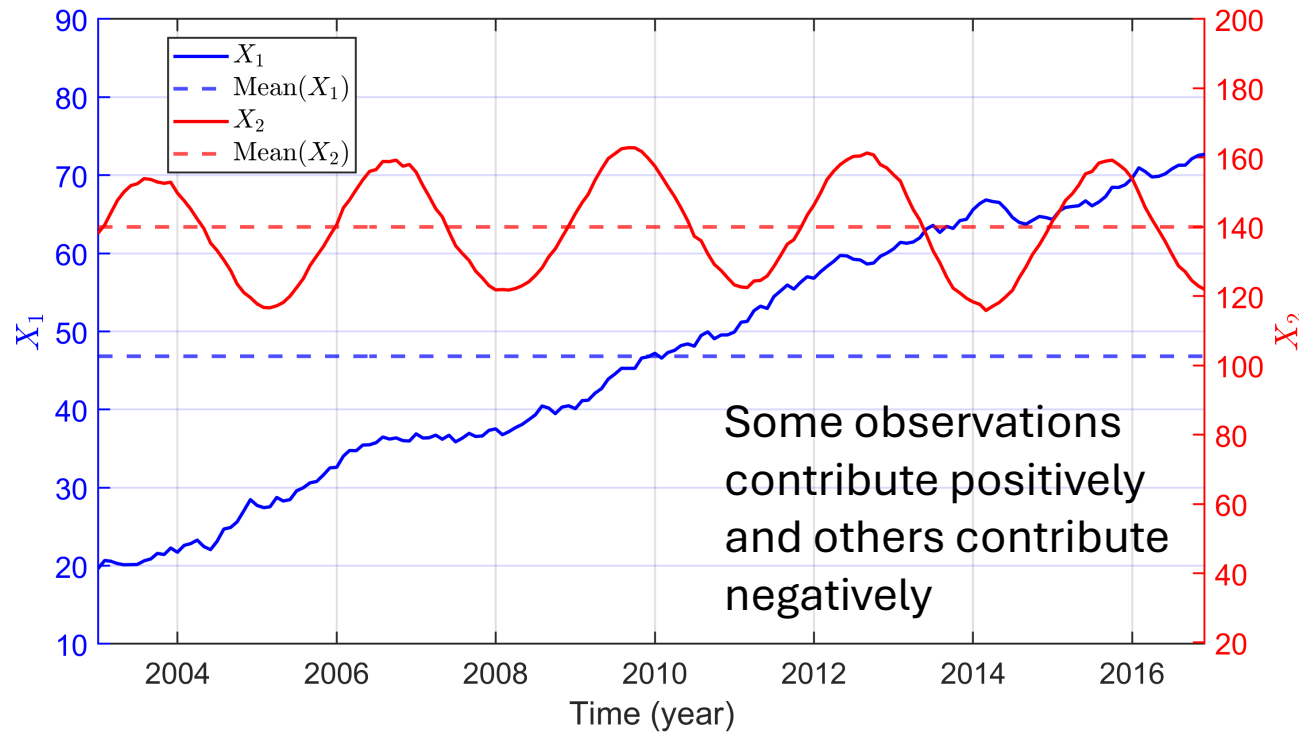
Covariance is **positive** when two variables tend to be above their means or below their means at the same time (i.e., **they show similar behavior**).

Negative covariance: Opposite behavior



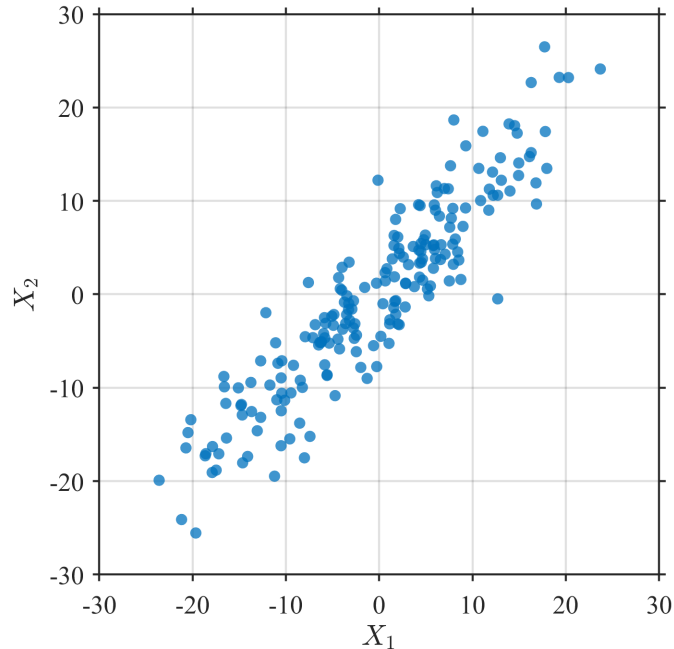
Covariance is **negative** when one variable tends to be above its mean while the other is below its mean (i.e., **they show opposite behavior**).

Near-zero covariance: No consistent relationship

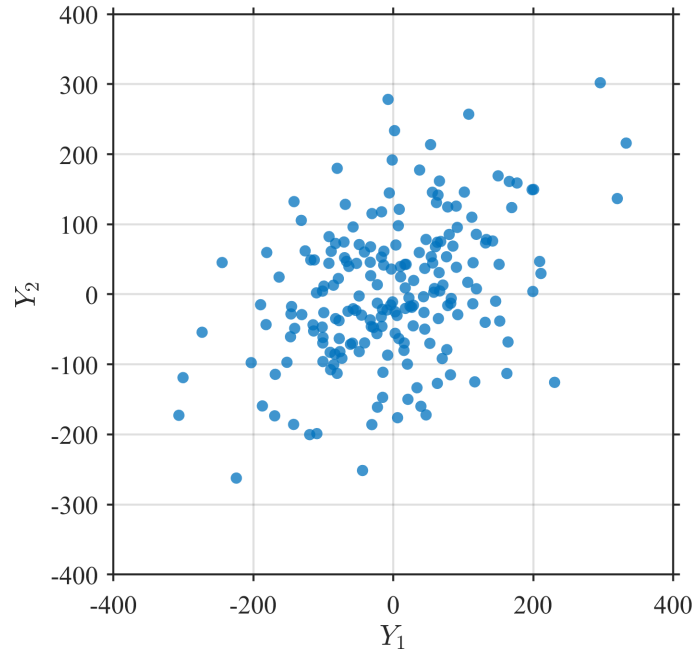


Positive and negative contributions **balance** each other, resulting in near-zero covariance. (i.e., **No consistent relationship**).

Covariance depends on the units of the variables



$$\text{Cov}(X_1, X_2) = 92.6$$



$$\text{Cov}(Y_1, Y_2) = 4234.1$$

Which pair is more strongly related?

The magnitude of covariance is difficult to interpret and compare.

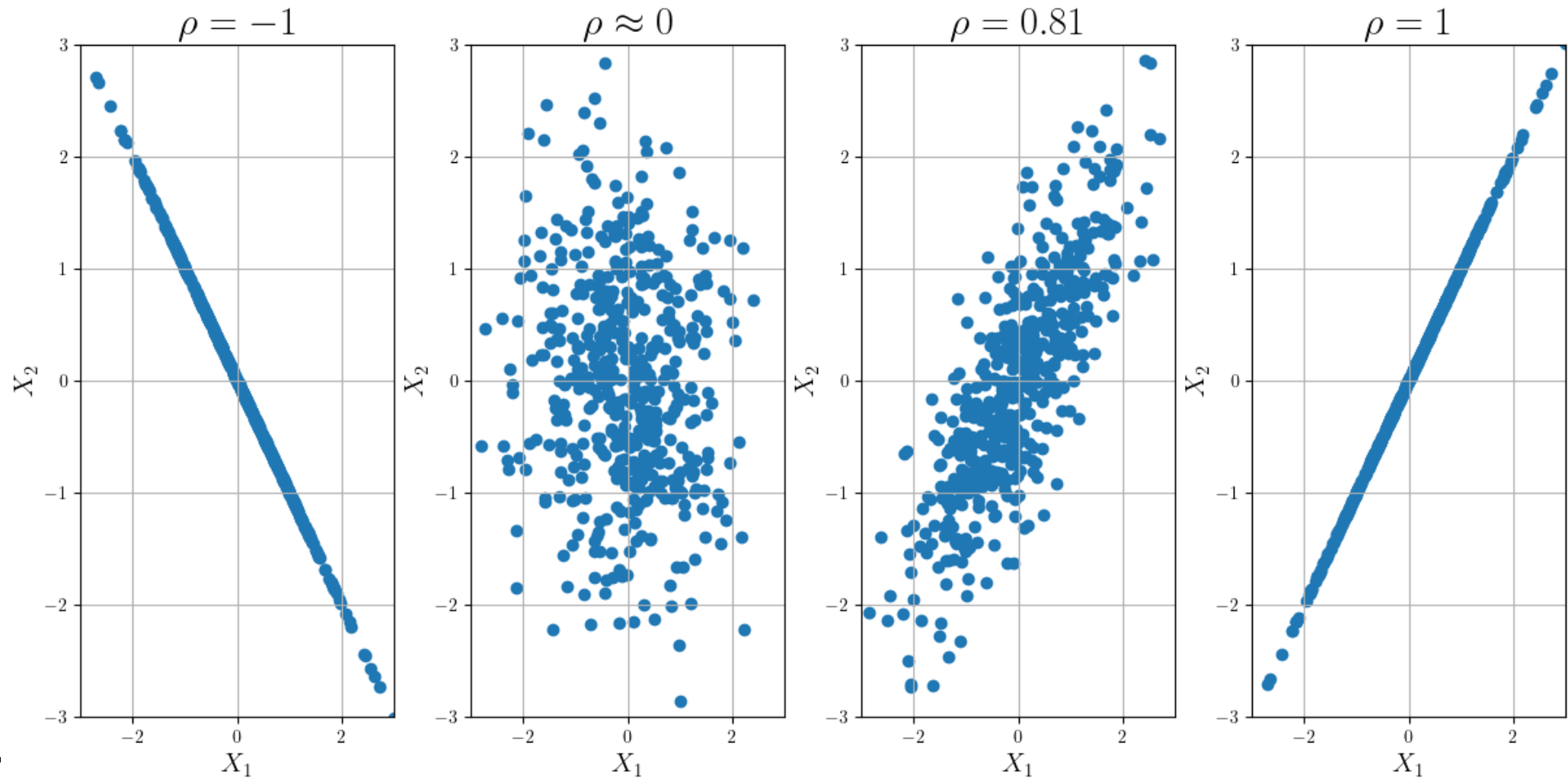
Correlation coefficient

Pearson's correlation coefficient: **standardizes** covariance by the variability of each variable.

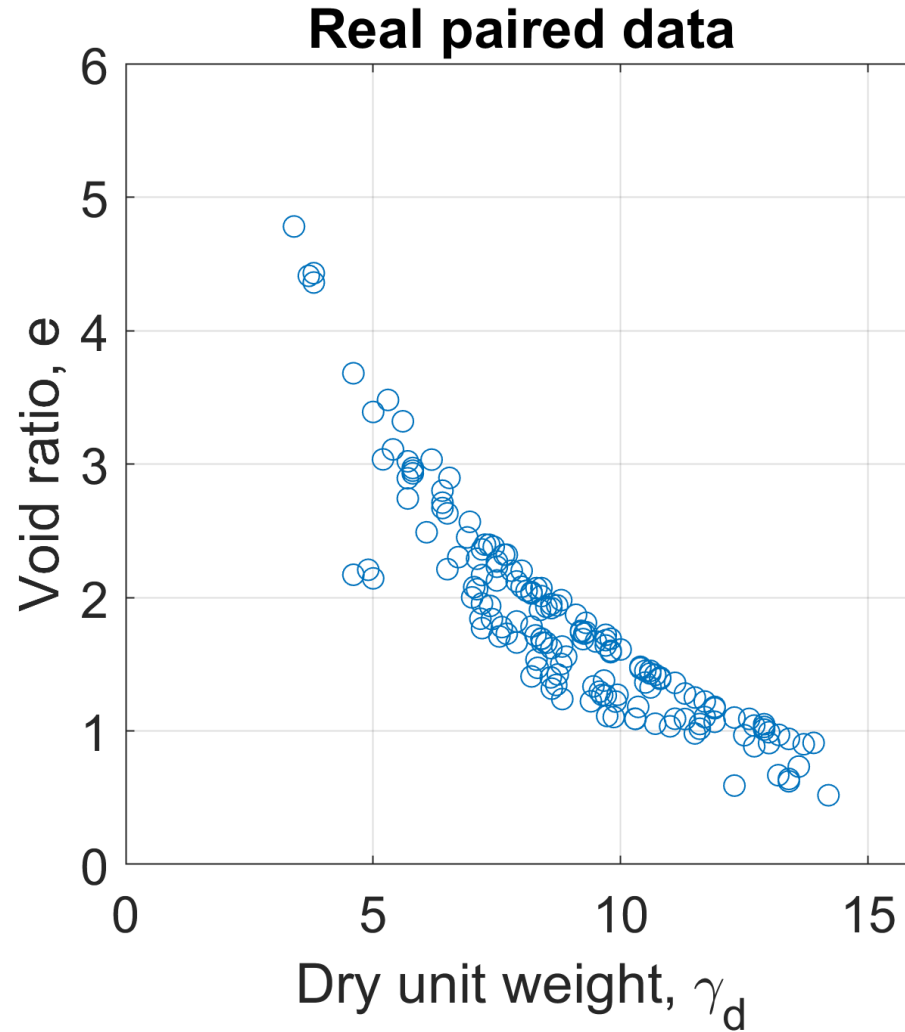
$$\rho = \frac{Cov(X_1, X_2)}{\sigma_{X_1} \sigma_{X_2}} = \frac{\sum_{i=1}^n (X_{1i} - \bar{X}_1)(X_{2i} - \bar{X}_2)}{\sqrt{\sum_{i=1}^n (X_{1i} - \bar{X}_1)^2 \sum_{i=1}^n (X_{2i} - \bar{X}_2)^2}}$$

ρ	Meaning
+1	Perfect positive linear relationship
0	No linear relationship
-1	Perfect negative linear relationship

Examples



How strongly are geotechnical variables related?



$\rho = -0.87$
*Strong **negative** linear
relationship*

Contents – Multivariate distributions

1. Measuring dependence (covariance/correlation)
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Why do we need a joint distribution?

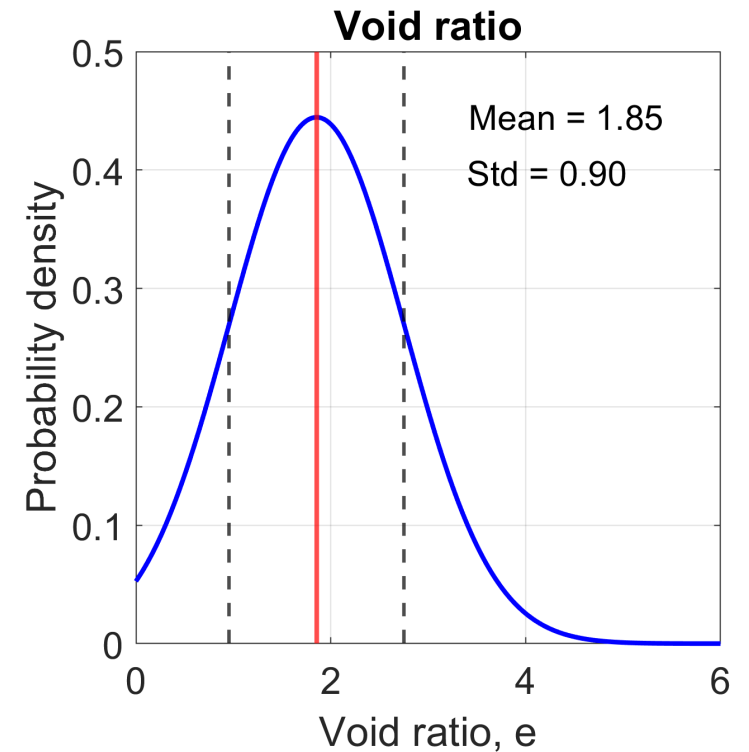
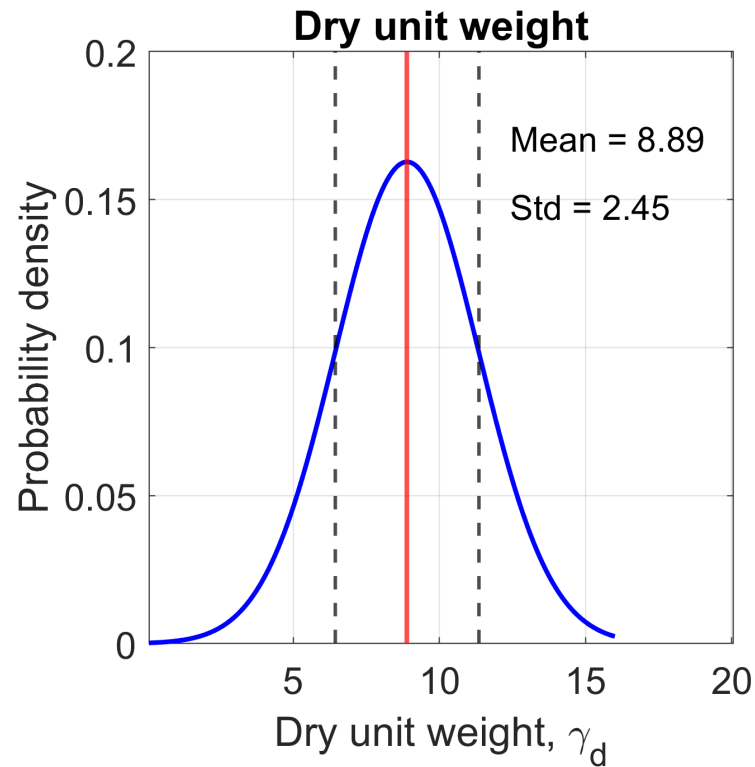
Marginal distributions

- Uncertainty in γ_d
- Uncertainty in e

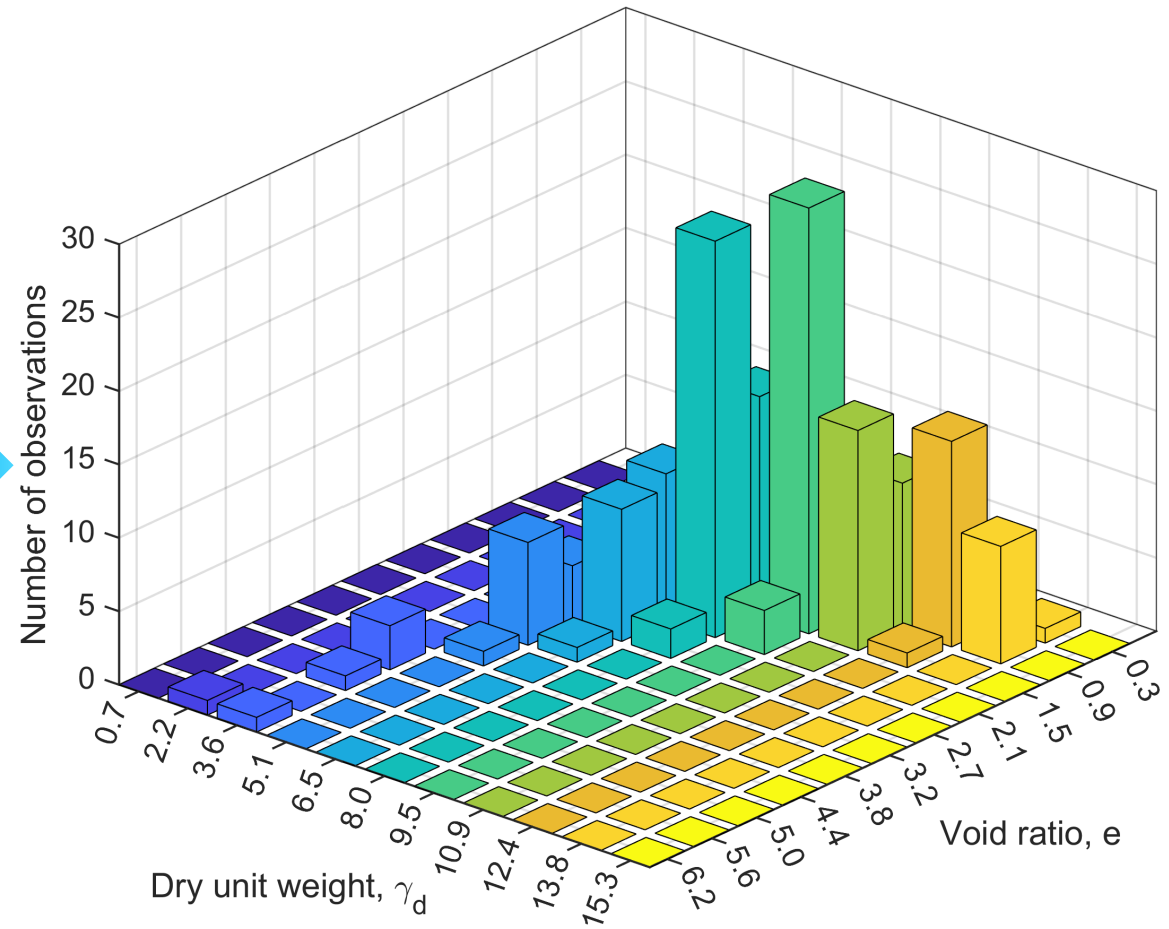
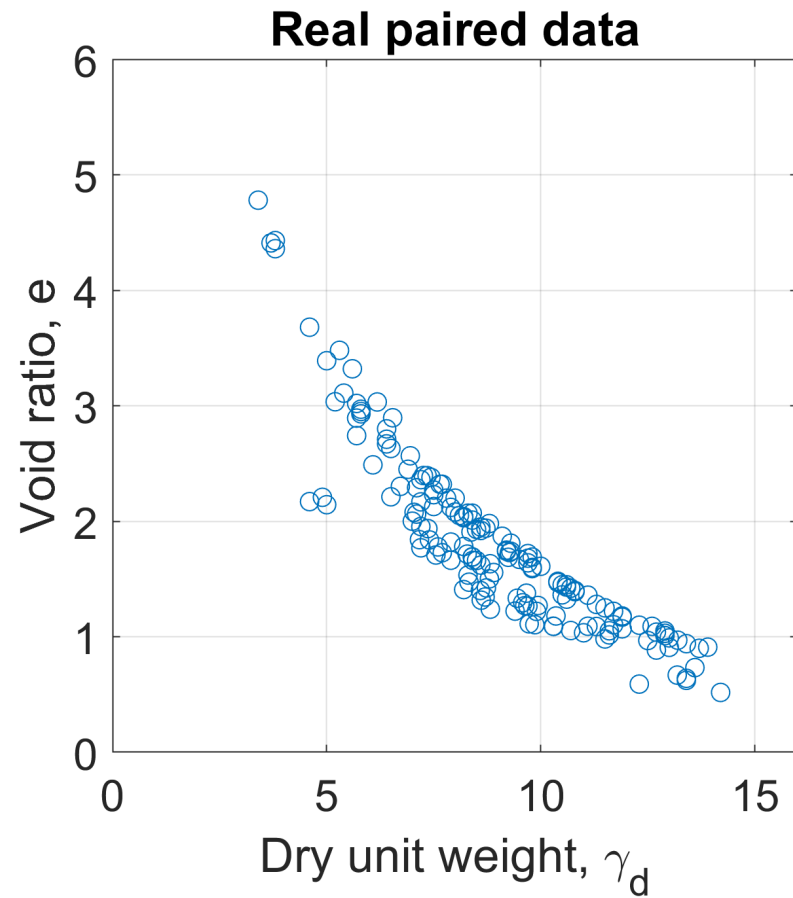
Two soils may have:

- same distribution of γ_d
- same distribution of e

but completely different **dependence structures**.



Joint histogram



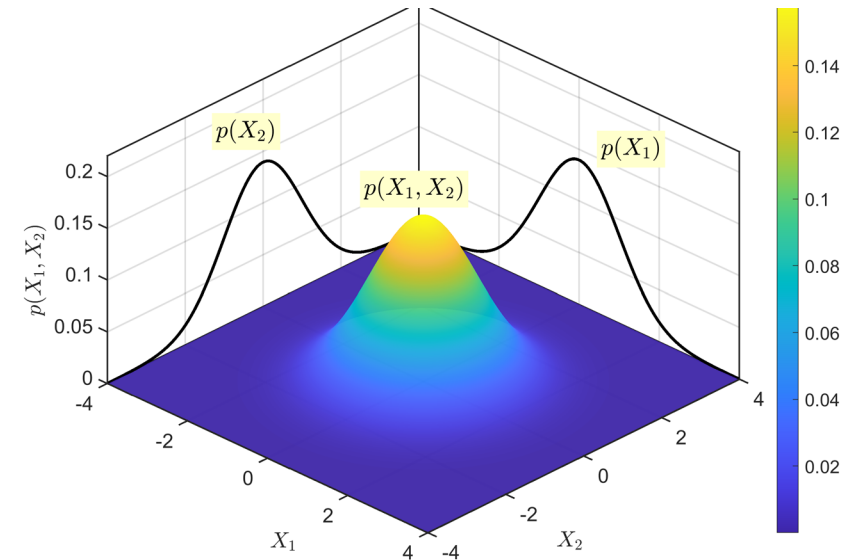
Joint probability distribution

Joint Probability Density Function (PDF)

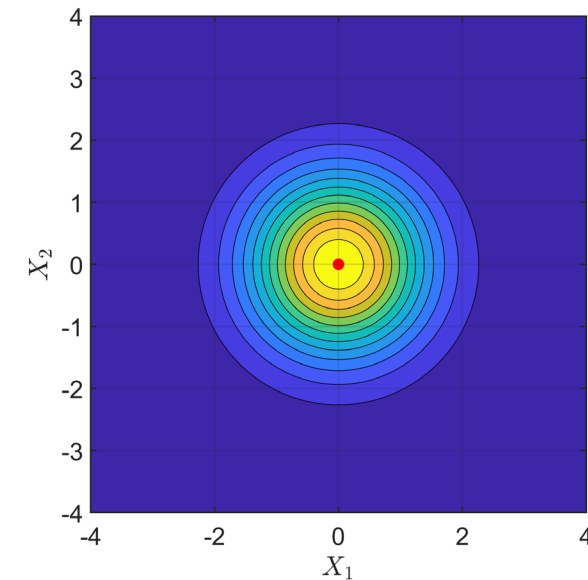
$$f_{X_1, X_2}(x_1, x_2)$$

Describes how likely different combinations of two variables

- High density → more likely
- Low density → less likely



3D surface



Contour plot

Contents – Multivariate distributions

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From univariate to bivariate Gaussian distributions

$X \sim N(\mu, \sigma^2)$ one variable $\rightarrow$ Gaussian distribution



$(X_1, X_2) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ two variables $\rightarrow$ Bivariate Gaussian distribution.

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \text{Mean vector} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \quad \text{Covariance matrix}$$

Bivariate Gaussian distribution equation

Univariate Gaussian

$$X \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right]$$

Normalization Standardized distance from the mean

Bivariate Gaussian

$$(X_1, X_2) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}} \exp \left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right] \quad (\text{matrix form})$$

Normalization Distance from the mean accounting for covariance

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right) \right]$$

No correlation $\rho = 0$

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2} \exp \left[-\frac{1}{2} \left(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right) \right]$$

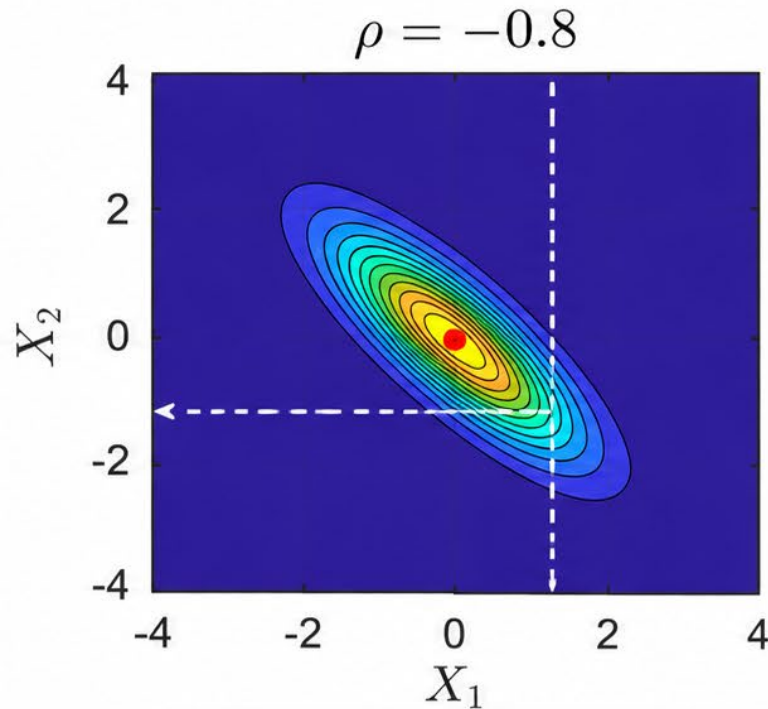
Notice that this can be written as

$$f_{X_1, X_2}(x_1, x_2) = \left[\frac{1}{\sqrt{2\pi}\sigma_1} \exp \left(-\frac{1}{2} \frac{(x_1 - \mu_1)^2}{\sigma_1^2} \right) \right] \left[\frac{1}{\sqrt{2\pi}\sigma_2} \exp \left(-\frac{1}{2} \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right) \right]$$

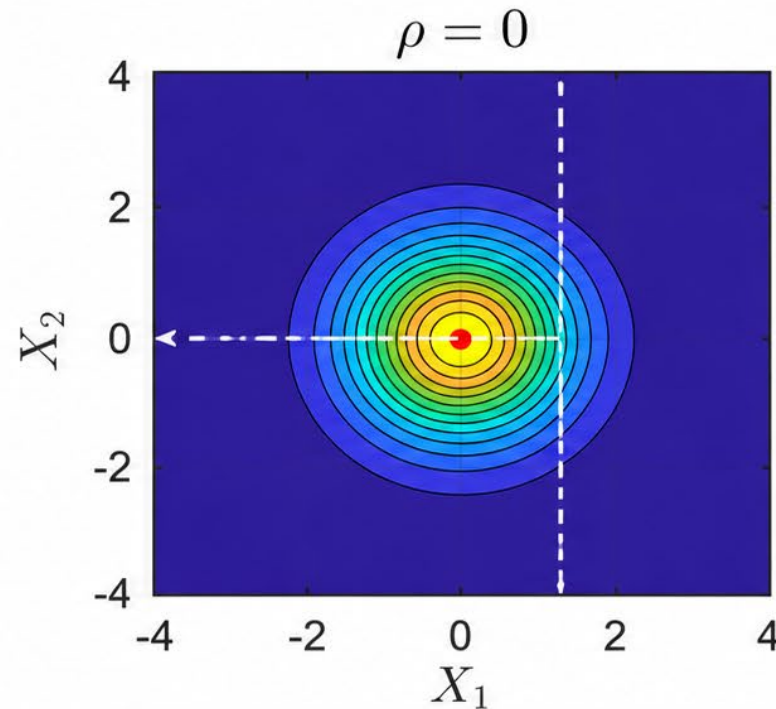
which is simply

$$\boxed{f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)}$$

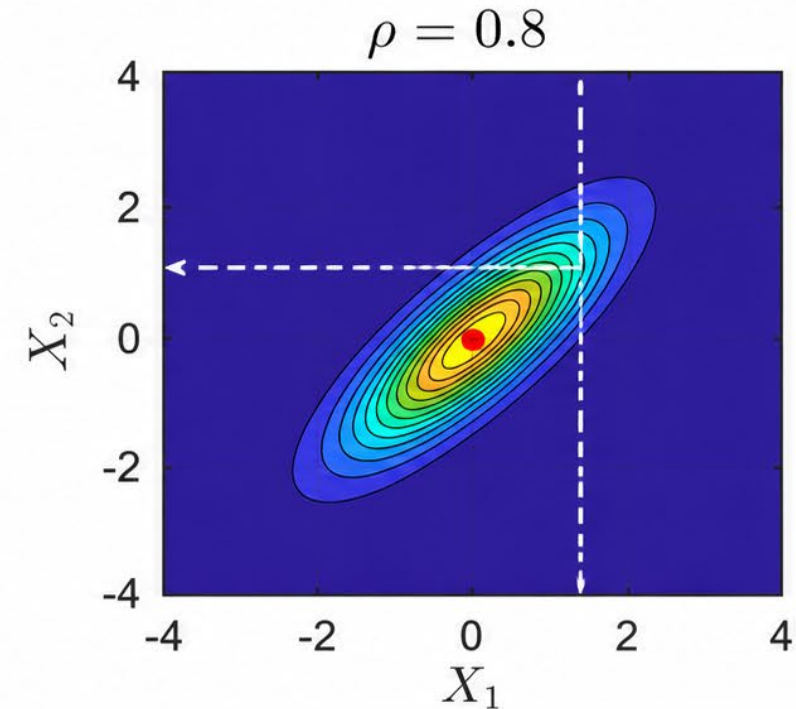
Effect of correlation



$\rho < 0$:
Larger X_1 typically comes
with lower X_2 .

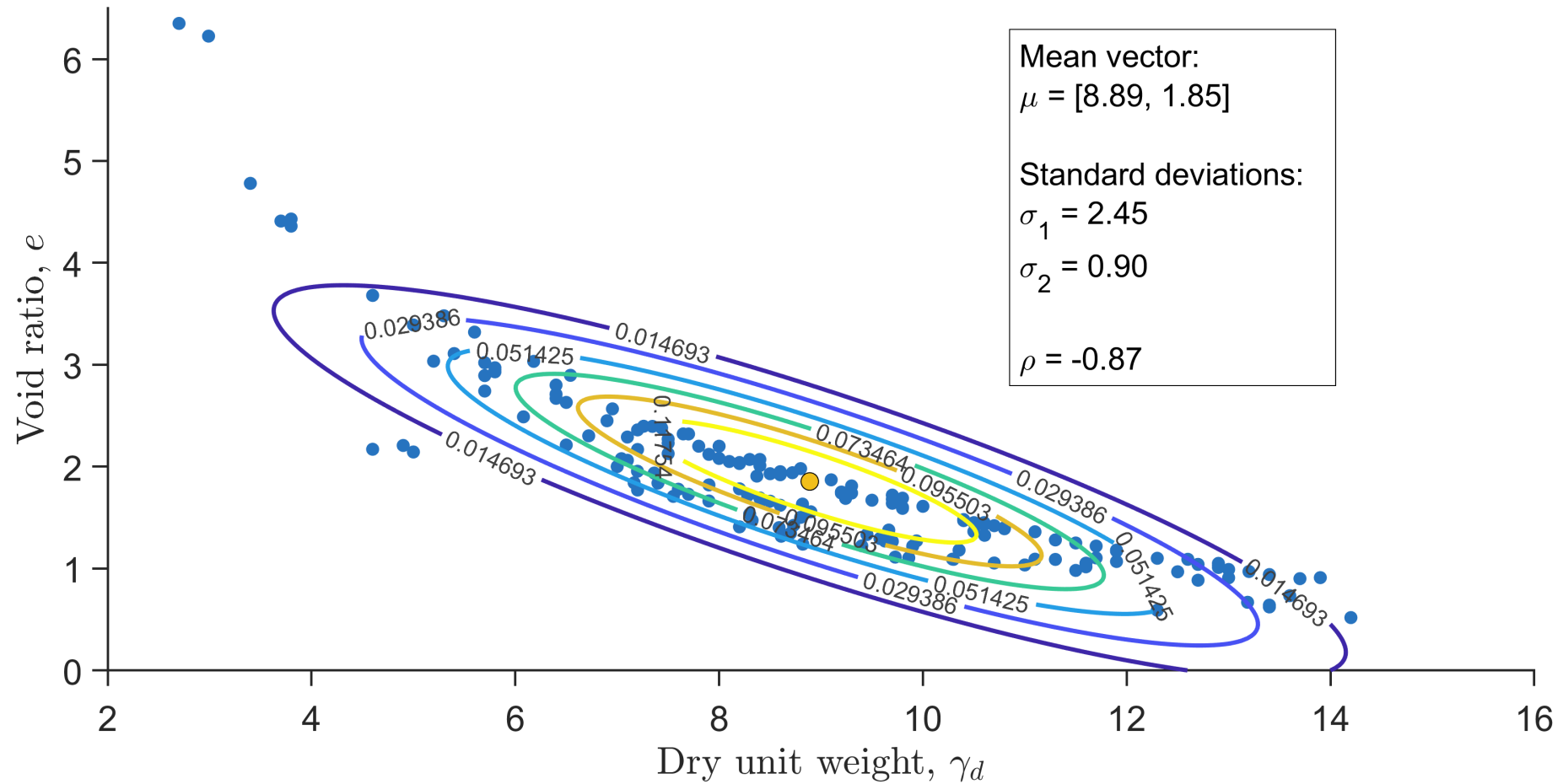


$\rho = 0$:
 X_1 and X_2 are not linearly related.
Larger X_1 does not change the typical X_2 .

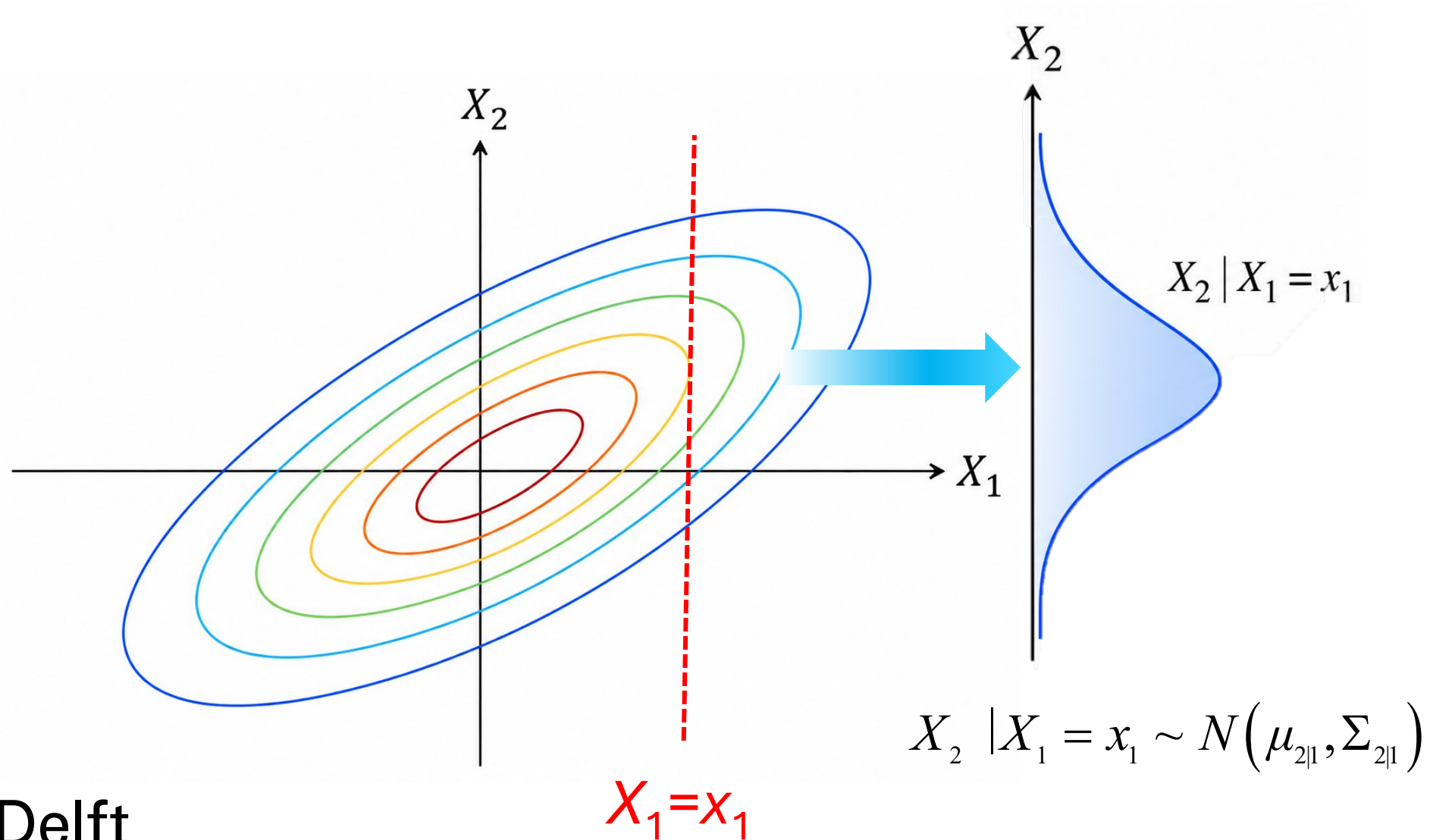


$\rho > 0$:
Larger X_1 typically comes
with higher X_2 .

Real geotechnical example



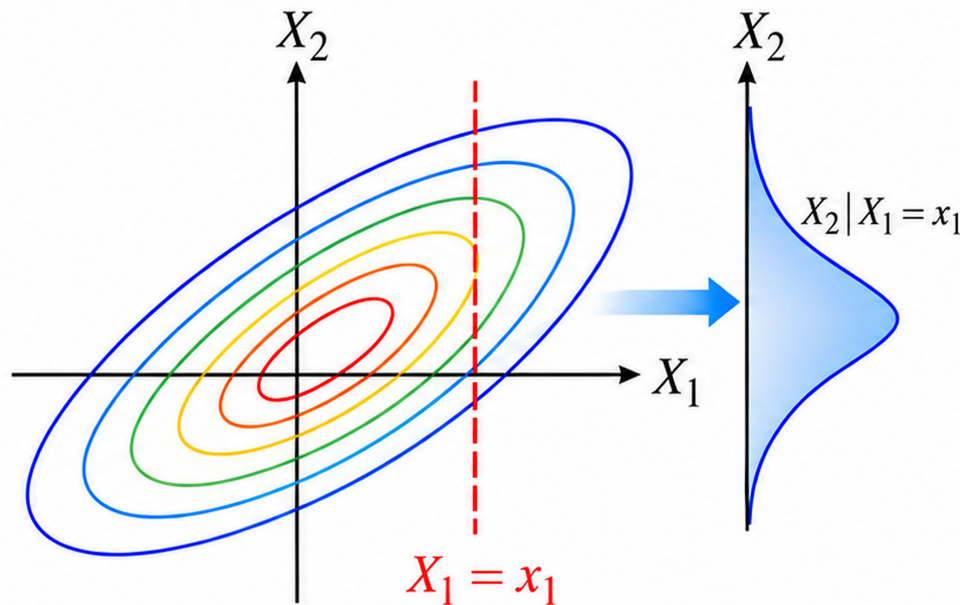
Conditional Gaussian distribution — Bivariate case



Conditional Gaussian distribution equation

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right)$$

$$\begin{aligned} \sigma_1^2 &= \text{Var}(X_1) & \rho &= \text{Corr}(X_1, X_2) \\ \sigma_2^2 &= \text{Var}(X_2) & (\text{so } \text{Cov}(X_1, X_2) &= \rho\sigma_1\sigma_2) \end{aligned}$$



Conditional distribution of X_2 given $X_1 = x_1$

$$X_2 | X_1 = x_1 \sim \mathcal{N}(\mu_{2|1}, \sigma_{2|1}^2)$$

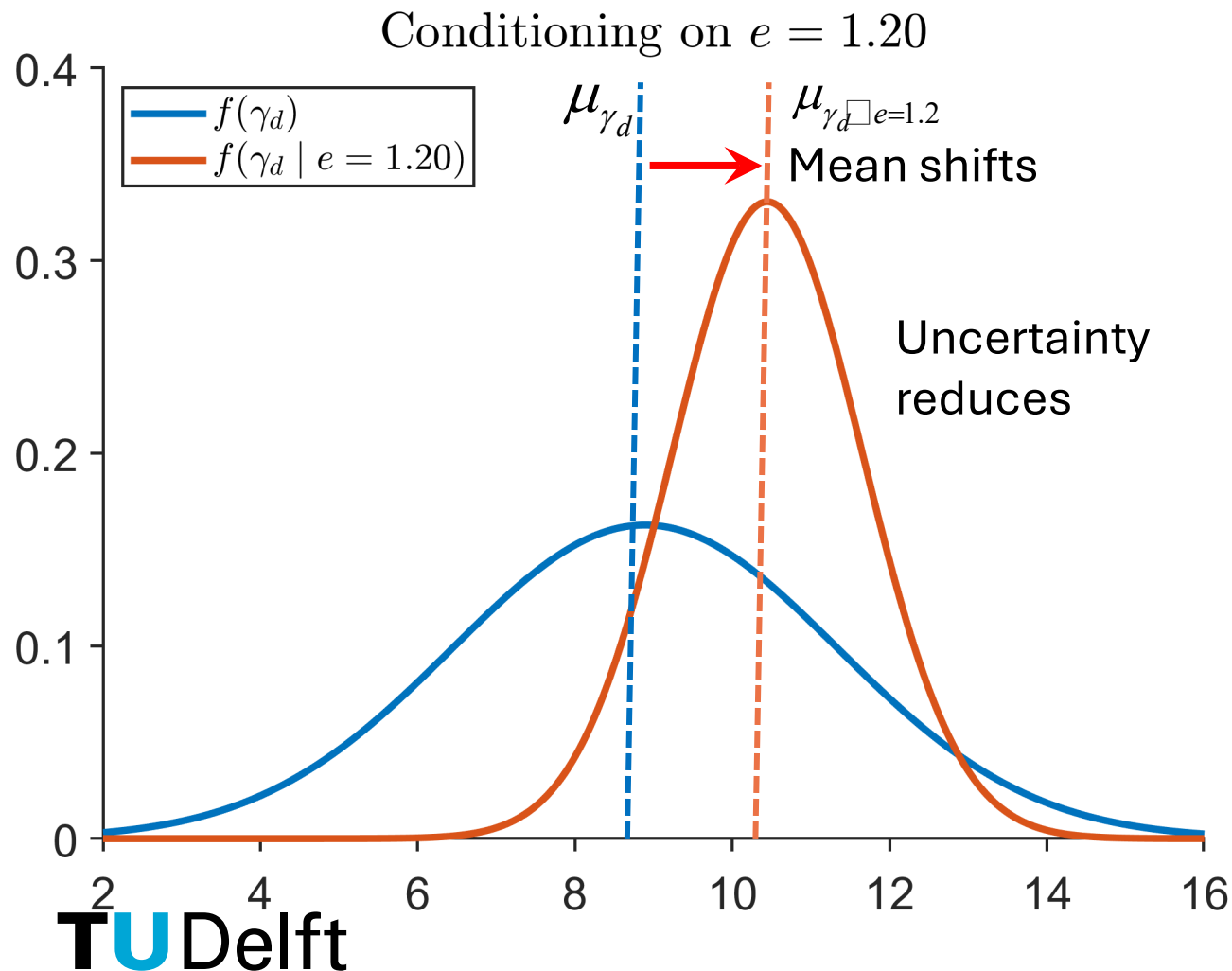
Conditional mean

$$\underbrace{\mu_{2|1}}_{\text{updated mean}} = \underbrace{\mu_2}_{\text{original mean}} + \underbrace{\rho \frac{\sigma_2}{\sigma_1} (x_1 - \mu_1)}_{\text{update from observing } X_1}$$

Conditional variance

$$\underbrace{\sigma_{2|1}^2}_{\text{remaining uncertainty}} = \underbrace{\sigma_2^2}_{\text{original uncertainty}} \underbrace{(1 - \rho^2)}_{\text{reduction due to correlation}}$$

Conditional Gaussian Distribution — Effect of Conditioning



Suppose we measure: $e = 1.20$

Question:

What can we infer about γ_d ?

$$\gamma_d \sim N(8.89, 2.45^2)$$



observe $e = 1.20$

$$\gamma_d | e = 1.20 \sim N(10.45, 1.21^2)$$

Knowing e provides information about γ_d because they are strongly correlated

Contents – Multivariate distributions

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Multivariate Gaussian distribution

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

$$\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\boldsymbol{\Sigma}|}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

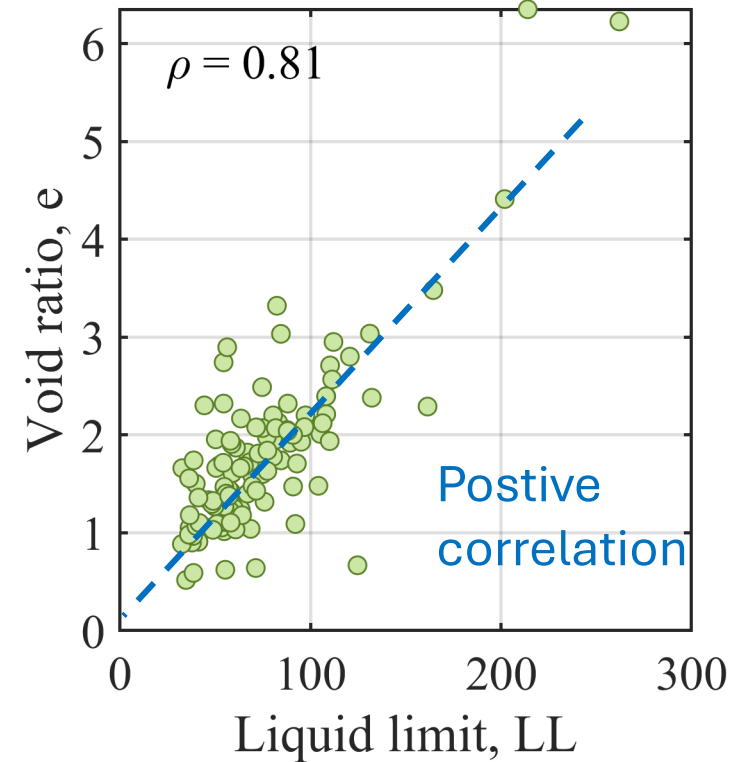
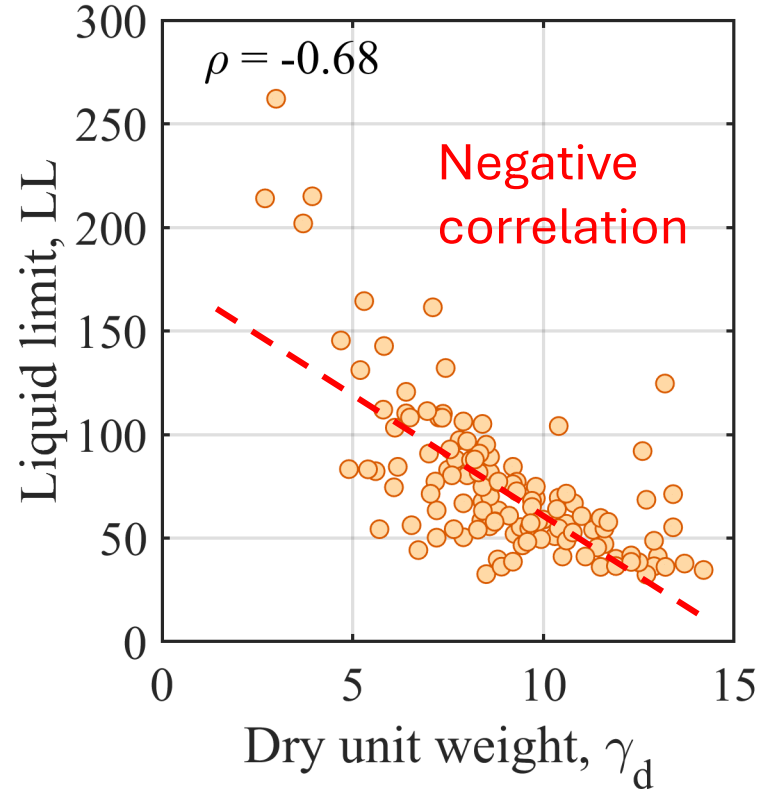
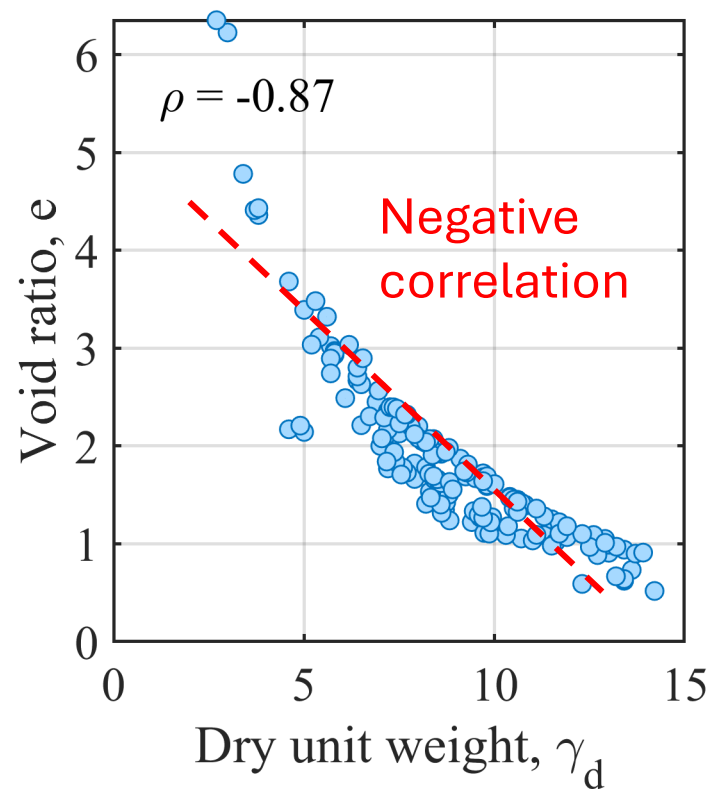
Mean vector

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho_{12} \sigma_1 \sigma_2 & \cdots & \rho_{1n} \sigma_1 \sigma_n \\ \rho_{21} \sigma_2 \sigma_1 & \sigma_2^2 & \cdots & \rho_{2n} \sigma_2 \sigma_n \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} \sigma_n \sigma_1 & \rho_{n2} \sigma_n \sigma_2 & \cdots & \sigma_n^2 \end{bmatrix}$$

Covariance matrix

The bivariate Gaussian extends the familiar normal distribution to two variables.

Multiple geotechnical variables



Multivariate analysis is needed to describe dependence among multiple geotechnical variables simultaneously.

Covariance matrix

Symmetric matrix: $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$

$$X = \begin{bmatrix} \gamma_d \\ e \\ LL \end{bmatrix} = \begin{bmatrix} \text{Dry unit weight} \\ \text{Void ratio} \\ \text{Liquid limit} \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_{\gamma_d}^2 & \rho_{\gamma_d,e} \sigma_{\gamma_d} \sigma_e & \rho_{\gamma_d,LL} \sigma_{\gamma_d} \sigma_{LL} \\ \rho_{\gamma_d,e} \sigma_{\gamma_d} \sigma_e & \sigma_e^2 & \rho_{e,LL} \sigma_e \sigma_{LL} \\ \rho_{\gamma_d,LL} \sigma_{\gamma_d} \sigma_{LL} & \rho_{e,LL} \sigma_e \sigma_{LL} & \sigma_{LL}^2 \end{bmatrix} \begin{matrix} \gamma_d \\ e \\ LL \end{matrix}$$

Diagonal entries
are variances

$$\Sigma = \begin{bmatrix} 7.2 & -1.9 & -62.5 \\ -1.9 & 0.8 & 25.6 \\ -62.5 & 25.6 & 1395.1 \end{bmatrix}$$

Correlation matrix

$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sigma_i \sigma_j}$$

$$R = \begin{matrix} & \begin{matrix} \gamma_d & e & LL \end{matrix} \\ \begin{bmatrix} 1 & -0.87 & -0.68 \\ -0.87 & 1 & 0.81 \\ -0.68 & 0.81 & 1 \end{bmatrix} & \begin{matrix} \gamma_d \\ e \\ LL \end{matrix} \end{matrix}$$

Conditional Gaussian Distribution

$$\mathbf{X} = \begin{bmatrix} \gamma_d \\ e \\ LL \end{bmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$f(\gamma_d, e, LL)$$

Describes our uncertainty before any observations are available

Suppose we measure: $LL = 120$

Question:

What can we infer about e and γ_d ?

$$f(\gamma_d, e, LL) \rightarrow f(\gamma_d, e \mid LL = 120)$$

Observations should **update** our predictions

Conditional Gaussian distribution equation

1 Split variables into two groups

$$X = \begin{bmatrix} LL \\ \gamma_d \\ e \end{bmatrix} \quad X_1 = \begin{bmatrix} LL \end{bmatrix} \quad X_2 = \begin{bmatrix} \gamma_d \\ e \end{bmatrix}$$

observed **unknown**

2 Partition the covariance matrix the same way

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

Σ_{11} **uncertainty of X_1**
 Σ_{12}, Σ_{21} **links between X_1 and X_2**
 Σ_{22} **uncertainty of X_2**

3 Condition on the observed variables and update the unknown variables

Observe $X_1 = x_1$  **Update** X_2

$$X_2 \mid X_1 = x_1 \sim N(\mu_{2|1}, \Sigma_{2|1})$$

Conditional mean — updated prediction

$$\mu_{2|1} = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$$

original mean + correction from observing X_1

Conditional covariance — reduced uncertainty

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$$

original uncertainty – uncertainty explained by X_1

Partitioning the covariance matrix

Geotechnical covariance matrix

	LL	γ_d	e
LL	1395.1	-62.5	25.6
γ_d	-62.5	7.2	-1.9
e	25.6	-1.9	0.8

Same matrix, viewed as four blocks

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

What do the blocks mean?

$$\Sigma_{11} = Var(LL)$$

Variance of the
observed variable

$$\Sigma_{21} = Cov([\gamma_d, e], LL)$$

How X_2 is related to
the observed X_1

$$\Sigma_{22} = Cov([\gamma_d, e])$$

Uncertainty and
covariance within X_2

Conditional mean and covariance: $\mu_{2|1} = \mu_2 + \Sigma_{21} \Sigma_{11}^{-1} (x_1 - \mu_1)$ $\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$

Conditional mean — updating predictions with observed LL

Conditional mean

$$\mu_{2|1}$$

=

Original prediction

$$\begin{bmatrix} \mu_{\gamma_d} \\ \mu_e \end{bmatrix}$$

+

Update coefficients

$$\begin{bmatrix} -0.0448 \\ 0.0184 \end{bmatrix}$$

New information

$$(LL - \mu_{LL})$$

updated prediction = original prediction + correction from the observation

Observation

$$LL = 120 > \mu_{LL}$$

LL is above its mean



Relationship with LL

γ_d : negative

e : positive

Updated prediction

$$\mu_{\gamma_d | LL} \downarrow$$

$$\mu_{e | LL} \uparrow$$

One observation updates several correlated soil properties simultaneously

Conditional covariance — reduced uncertainty after observing LL

Conditional covariance

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$$

uncertainty explained by observed LL

Before observing LL

$$\Sigma_{22} = \begin{bmatrix} 7.2 & -1.9 \\ -1.9 & 0.8 \end{bmatrix}$$

Diagonal entries = variances

$$\text{Var}(\gamma_d) = 7.2 \quad \text{Var}(e) = 0.8$$

observe LL



After observing LL

$$\Sigma_{2|1} = \begin{bmatrix} 4.40 & -0.75 \\ -0.75 & 0.33 \end{bmatrix}$$

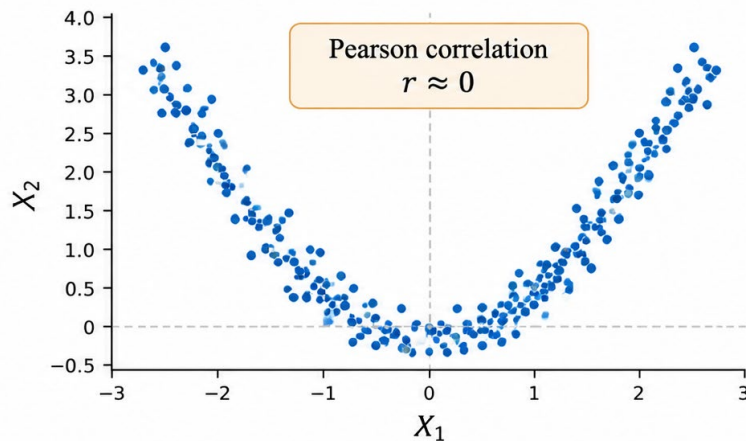
Updated variances

$$\text{Var}(\gamma_d | LL) = 4.40 \quad \text{Var}(e | LL) = 0.33$$

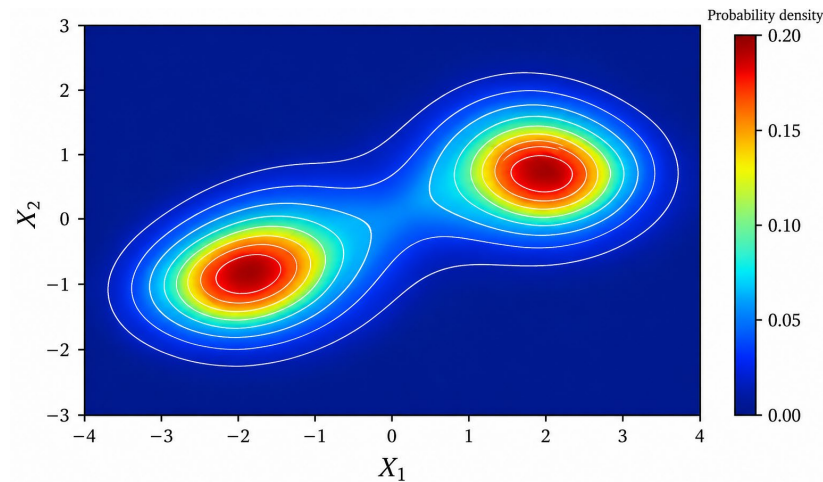
Observing LL provides information about both γ_d and e , so their uncertainty decreases.

Beyond the multivariate Gaussian model

Nonlinear dependence



Non-Gaussian joint shapes



Other possibilities

Transformations

Copula models

**Non-Gaussian
multivariate models**

Mixture models

Data-driven models

Wrap up

- Dependence is quantified by covariance and correlation
- Joint probability distributions describes the uncertainty of multiple variables simultaneously

- Multivariate Gaussian distributions provide a compact model

$$\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- Observations reduce uncertainty: Conditioning updates predictions using observed information.

What's next?

- There is more in the textbook!
- Wednesday workshop: Bivariate Gaussian distributions with CPT data
- Friday group assignment: Gaussian Soil

And enjoy the journey!

Derivation of the conditional bivariate Gaussian (1/2)

For two jointly Gaussian variables,

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right) \Rightarrow f(x_1, x_2) \propto \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1 - \mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2}\right)\right]$$

For a given $X_1 = x_1$,

$$f(x_2 | x_1) = \frac{f(x_1, x_2)}{f_{X_1}(x_1)} \propto f(x_1, x_2) \Rightarrow f(x_2 | x_1) \propto \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_2 - \mu_2)^2}{\sigma_2^2} - 2\rho\frac{(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1\sigma_2}\right]\right\}$$

Since x_1 is fixed, keep only the terms involving x_2

Key idea: Terms depending only on x_1 are constants and do not affect the shape of the conditional PDF

Derivation of the conditional bivariate Gaussian (2/2)

Complete the square with respect to x_2 :

$$(x_2 - \mu_2)^2 - 2\rho \frac{\sigma_2}{\sigma_1} (x_1 - \mu_1)(x_2 - \mu_2) \propto \left[x_2 - \mu_2 - \rho \frac{\sigma_2}{\sigma_1} (x_1 - \mu_1) \right]^2$$

Therefore,

$$f(x_2 | x_1) \propto \exp \left[- \frac{\left[x_2 - \mu_2 - \rho \frac{\sigma_2}{\sigma_1} (x_1 - \mu_1) \right]^2}{2\sigma_2^2(1-\rho^2)} \right] \quad X_2 | X_1 = x_1 \sim N \left(\underbrace{\mu_2 + \rho \frac{\sigma_2}{\sigma_1} (x_1 - \mu_1)}_{\mu_{2|1}}, \underbrace{\sigma_2^2(1-\rho^2)}_{\Sigma_{2|1}} \right)$$